

HYDROGENATED NBR (HNBR) SYNTHESIS AND POST-PROCESSING PATENT LANDSCAPE

1. ABSTRACT

This report synthesizes three primary US patents relevant to hydrogenated nitrile rubber (HNBR): one focused on removal of ruthenium catalyst residues from optionally hydrogenated nitrile rubber, and two focused on producing low molecular weight HNBR with narrower molecular weight distributions. Across the set, the target material is consistently defined as nitrile rubber derived from conjugated dienes and α,β -unsaturated nitriles that is selectively hydrogenated to improve heat, oil, and chemical resistance while retaining processable elastomeric properties. The patents collectively address two major manufacturing challenges: controlling molecular weight/Mooney viscosity during hydrogenation and removing residual transition-metal catalysts after synthesis.

The first patent discloses purification of optionally hydrogenated nitrile rubber solutions using functionalized macroreticular ion-exchange resins, emphasizing low ruthenium content in the recovered polymer. It is relevant to downstream HNBR quality because catalyst residue control is critical for color, stability, and compatibility with subsequent processing. The second patent introduces a two-step route in which nitrile rubber is first subjected to metathesis in the presence of ruthenium or osmium compounds, optionally with a co-olefin and solvent, and then hydrogenated to yield lower-MW HNBR with reduced polydispersity. The third patent advances this concept further by performing simultaneous metathesis and hydrogenation in a single step using specific ruthenium/osmium catalyst systems, enabling low residual double bond content and lower Mooney viscosity.

Major trends across the landscape include use of homogeneous transition-metal catalysts, co-olefin-assisted chain scission, and explicit control of molecular weight, Mooney viscosity, and residual unsaturation. Process conditions disclosed in the examples include solution-phase operation, elevated temperatures, hydrogen pressure, catalyst loading in phr, and post-reaction purification by filtration, evaporation, drying, or ion exchange. Overall, the evidence points to a technical direction centered on controlled depolymerization/hydrogenation chemistry followed by rigorous catalyst-removal strategies to produce commercially useful HNBR grades for seals, hoses, and other demanding elastomer applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis itself; it discloses purification of optionally hydrogenated nitrile rubber solutions by contacting them with a functionalized macroreticular ion-exchange resin to remove ruthenium-containing catalyst residues. The recovered rubber is then isolated by filtration, solvent evaporation, and drying. - Ion exchange resin loading: 0.5 g — Example 2A - Rubber solution charge: 180 g — Example 2A - Reaction temperature: ca. 100° C. — Example 2A - Reaction time: 66 hours — Example 2A - Drying temperature: 60° C. — Example 2A - Initial Ru content: 15 ppm — Table 2 - Inventive Ru content after treatment: 10 ppm — Table 2 - Ru removal: 33% — Table 2 - Average particle diameter: 0.2 to 2.5 mm dry basis — Claim 1 - Ruthenium content in solution: 5 to 1000 ppm — Claim 2 - Ruthenium content in solution: 5 to 500 ppm — Claim 3 - Polymer concentration in solution: 0.5 to 30 wt.% — Claim 4

Relevant Experimental Evidence - Example 2A: Functionalized ion-exchange resin reduced Ru content in nitrile rubber solution from 15 ppm to 10 ppm. - Example 2B: Untreated control rubber was recovered for comparison of initial Ru content. - Example 2: Table 2 quantified 33% Ru removal after resin treatment.

Technical Relevance This patent is relevant because catalyst residue removal is a key finishing step for HNBR made by homogeneous hydrogenation. Low ruthenium content improves product purity and supports downstream use in demanding elastomer applications.

Patent 2

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step route in which nitrile rubber is first subjected to metathesis in the presence of ruthenium or osmium compounds, optionally with a co-olefin and solvent, and the resulting material is then hydrogenated to HNBR. The process is designed to lower molecular weight and narrow molecular weight distribution while retaining the HNBR backbone chemistry. - Target molecular weight: 30,000 to 250,000 — Claim 1 - Target Mooney viscosity: 3 to 50 (ML 1+4 @ 100° C.) — Claim 1 - Target polydispersity index: less than 2.5 — Claim 1 - Rubber concentration: 15 wt.-% solid — Example 1 - Rubber concentration: 6 wt.-% solid — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - Co-olefin pressure: 500 psi — Table 1 - 1-hexene amount: 150 g — Example 4 / Table 1 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity: 73 — Therban® A3407 substrate - Substrate Mn: 85,000 — Table 2 - Substrate Mw: 296,000 — Table 2 - Substrate Mz: 939,000 — Table 2 - Substrate PDI: 3.50 — Table 2 - Experiment 1 Mn: 73,000 — Table 2 - Experiment 1 Mw: 189,000 — Table 2 - Experiment 1 Mz: 441,000 — Table 2 - Experiment 1 PDI: 2.59 — Table 2 - Experiment 1 Mooney viscosity: 43 — Table 2 - Experiment 2 Mn: 60,000 — Table 2 - Experiment 2 Mw: 136,000 — Table 2 - Experiment 2 Mz: 277,000 — Table 2 - Experiment 2 PDI: 2.27 — Table 2 - Experiment 2 Mooney viscosity: 28 — Table 2 - Experiment 3 Mn: 31,000 — Table 2 - Experiment 3 Mw: 59,000 — Table 2 - Experiment 3 Mz: 98,000 — Table 2 - Experiment 3 PDI: 1.90 — Table 2 - Experiment 3 Mooney viscosity: 3 — Table 2 - Experiment 4 Mn: 55,000 — Table 2 - Experiment 4 Mw: 111,000 — Table 2 - Experiment 4 Mz: 1,197,000 — Table 2 - Experiment 4 PDI: 2.02 — Table 2 - Experiment 4 Mooney viscosity: 31 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis/hydrogenation reduced molecular weight and Mooney viscosity relative to the substrate. - Example 2: Higher catalyst loading further reduced Mw and Mooney viscosity. - Example 3: Lower solid content produced the lowest Mw, PDI, and Mooney viscosity among the ethylene runs. - Example 4: 1-hexene co-olefin also enabled low-MW HNBR formation with controlled PDI. - Example 4: Table 2 showed up to a fivefold reduction in Mw and narrowing of PDI to 1.90.

Technical Relevance This patent directly advances HNBR synthesis by combining metathesis and hydrogenation to reduce molecular weight and improve processability. It is especially relevant for producing low-viscosity HNBR grades with controlled residual unsaturation.

Patent 3

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a single-step process for producing low molecular weight HNBR by simultaneously subjecting nitrile rubber to hydrogenation and metathesis in the presence of hydrogen, optionally a co-olefin, and a ruthenium- or osmium-based catalyst of formula (I). The process is intended to deliver lower molecular weight, narrower molecular weight distribution, and low residual double bond content in one operation. - Target molecular weight: 20,000 to 250,000 — Description - Target Mooney viscosity: 1 to 50 (ML 1+4 @ 100° C.) — Description - Target polydispersity index: less than 3.0 — Description - Hydrogenation catalyst amount: 0.041 phr — Table 1, HNBR 1 - Hydrogenation catalyst amount: 0.045 phr — Table 1, HNBR 2 - Hydrogenation catalyst amount: 0.060 phr — Table 1, HNBR 3 - Metal content: 65.6 ppm — Table 1 - Co-catalyst: NIL — Table 1, HNBR 1 - Co-catalyst: NIL — Table 1, HNBR 2 - Co-catalyst: Triphenylphosphine — Table 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Table 1, HNBR 3 - Reaction time: 60 min — Table 2 - RDB after 60 min: 2.5% — Table 2, HNBR 1 - RDB after 60 min: 7.0% — Table 2, HNBR 2 - RDB after 60 min: 15.4% — Table 2, HNBR 3 - Reaction time: 120 min — Table 2 - RDB after 120 min: 0.6% — Table 2, HNBR 1 - RDB after 120 min: 4.5% — Table 2, HNBR 2 - RDB after 120 min: 3.1% — Table 2, HNBR 3 - Reaction time: 180 min — Table 2 - RDB after 180 min: 0.5% — Table 2, HNBR 1 - RDB after 180 min: 3.9% — Table 2, HNBR 2 - RDB after 180 min: 1.3% — Table 2, HNBR 3 - Reaction time: 240 min — Table 2 - RDB after 240 min: 0.4% — Table 2, HNBR 1 - RDB after 240 min: 3.5% — Table 2, HNBR 2 - RDB after 240 min: 0.7% — Table 2, HNBR 3 - Mw after 240 min: 113,371 g/mol — Table 3, HNBR 1 - Mn after 240 min: 48,352 g/mol — Table 3, HNBR 1 - PDI after 240 min: 2.34 — Table 3, HNBR 1 - Mooney viscosity after 240 min: 33.3 — Table 3, HNBR 1 - Mw after 240 min: 133,758 g/mol — Table 3, HNBR 2 - Mn after 240 min: 54,595 g/mol — Table 3, HNBR 2 - PDI after 240 min: 2.45 — Table 3, HNBR 2 - Mooney viscosity after 240 min: 41.3 — Table 3, HNBR 2 - Mw after 240 min: 195,304 g/mol — Table 3, HNBR 3 - Mn after 240 min: 64,130 g/mol — Table 3, HNBR 3 - PDI after 240 min: 3.05 — Table 3, HNBR 3 - Mooney viscosity after 240 min: 69.3 — Table 3, HNBR 3

Relevant Experimental Evidence - Example 1: Hoveyda and Arlt catalyst systems achieved faster RDB reduction than Wilkinson's catalyst. - Example 1: HNBR 1 reached 0.4% RDB at 240 min with Mooney viscosity 33.3. - Example 1: HNBR 3 retained higher RDB and much higher Mooney viscosity, showing inferior performance. - Example 1: The inventive catalysts reduced molecular weight while hydrogenating the polymer. - Example 1: Table 3 confirmed narrower PDI for HNBR 1 and HNBR 2 than for the industrial benchmark.

Technical Relevance This patent is highly relevant because it directly teaches simultaneous hydrogenation/metathesis of NBR to make low-MW HNBR in a single step. It provides practical catalyst and process guidance for controlling residual unsaturation, viscosity, and molecular weight.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2 and US9150669B2 both target low molecular weight HNBR, but US7919563B2 uses a two-step metathesis-then-hydrogenation sequence, whereas US9150669B2 performs simultaneous metathesis and hydrogenation in one step.
- US7919563B2 explicitly uses co-olefins such as ethylene and 1-hexene, with 500 psi ethylene and 80° C. in the examples; US9150669B2 also allows optional co-olefin but the example data emphasize catalyst comparison rather than co-olefin identity.
- Catalyst systems differ: US7919563B2 claims Ru/Os compounds for metathesis, while US9150669B2 focuses on specific Ru/Os formula (I) catalysts and compares Hoveyda, Arlt, and Wilkinson's catalyst systems.
- US9150669B2 reports direct hydrogenation performance via residual double bond content over 60-240 min, reaching 0.4% RDB for HNBR 1 at 240 min; US7919563B2 reports final polymer properties such as Mw, Mn, PDI, and Mooney viscosity after metathesis/hydrogenation.
- US7846995B2 is not a synthesis patent but a purification patent; it complements the other two by addressing removal of ruthenium residues from optionally hydrogenated nitrile rubber solutions after synthesis.
- Across the set, solution-phase processing is common, but only US7846995B2 emphasizes post-reaction cleanup by ion exchange, while the other two emphasize molecular-weight control during formation of HNBR.

4. CONCLUSION

The patent set shows a coherent HNBR manufacturing strategy: first control chain length and residual unsaturation through metathesis-assisted hydrogenation, then remove catalyst residues to obtain a cleaner product. The strongest synthesis evidence supports solution-phase Ru/Os-catalyzed processes at elevated temperature, with optional co-olefins and hydrogen, to reach low Mw, low Mooney viscosity, and low RDB. Based on the examples, effective parameter windows include about 80° C., catalyst loadings in the 0.041-0.060 phr range for hydrogenation/metathesis systems, and sufficient reaction time to drive RDB below 1% in the best case. For product purification, functionalized macroreticular ion-exchange resins with particle diameters of 0.2 to 2.5 mm and treatment of rubber solutions containing 5 to 500 ppm Ru are supported by the evidence. Overall, the evidence favors a combined approach of controlled catalytic depolymerization/hydrogenation followed by rigorous metal-removal purification to produce commercially useful HNBR for seals, hoses, and other high-performance elastomer applications.

5. REFERENCES

- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>

- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>